Data Science Capstone Project Report: SpaceX Falcon 9 First Stage Landing Prediction

Author: Nursultan Raimzhanov **Affiliation: North American University** **Date: October 2026**

Project GitHub Repository URL (All Completed Notebooks & Python Code):
`https://github.com/theraimzhanov/testrepo`

1. Executive Summary

Summary of Methodologies: This capstone study establishes an end-to-end data science pipeline to predict whether the SpaceX Falcon 9 first-stage booster will successfully land. Raw mission data was gathered programmatically via the SpaceX REST API v4 and supplemented through web scraping Wikipedia Falcon 9 launch tables using `BeautifulSoup`. Data wrangling involved filtering out Falcon 1 flights, imputing missing payload mass values with the dataset mean (`6,123.5 kg`), mapping landing outcomes to a binary target label (`Class = 1` for success, `Class = 0` for failure), and applying one-hot encoding to categorical features. Exploratory Data Analysis (EDA) was conducted via Matplotlib/Seaborn visualizations and 10 structured SQLite queries. Interactive spatial analytics and real-time dashboards were built using Folium and Plotly Dash. Finally, four machine learning classification models (Logistic Regression, SVM, Decision Tree, and KNN) were trained on standardized features and tuned via 10-fold `GridSearchCV`.

Summary of All Results: Overall launch and landing reliability improved dramatically from 2013 (0% landing success) to 2020 (84%+ landing success). Orbits such as ES-L1, GEO, HEO, and SSO achieved a 100% landing success rate, whereas heavy Geostationary Transfer Orbit (GTO) missions had a 51.9% success rate. Spatial analysis proved that all launch pads are strategically positioned within `0.85 km` of coastlines and `>15 km` from major cities. In the Plotly Dash analysis, KSC LC-39A accounted for the largest share of successful launches (41.7%) and highest site success rate (76.9%), while the Falcon 9 FT booster achieved the highest success rate across payload ranges of `2,000–5,500 kg`. All four tuned classification models achieved an identical **83.33% accuracy** (15/18 correct predictions) on the unseen test dataset, with 12 True Positives, 3 True Negatives, 3 False Positives, and 0 False Negatives.

2. Introduction & Project Outline

Project Background & Context: SpaceX has revolutionized the commercial aerospace industry by offering Falcon 9 rocket launches at an advertised price of **$62 million**, compared to upward of **$165 million** charged by traditional aerospace providers. This dramatic cost reduction is driven primarily by SpaceX's ability to recover and reuse the expensive first-stage booster via ground-pad (RTLS) or autonomous drone-ship (ASDS) landings.

Core Problem Statement: A competing aerospace enterprise ("Space Y") intends to bid against SpaceX for commercial and government satellite launch contracts. Because the true marginal cost of a Falcon 9 launch depends directly on whether the first stage is recovered, accurately predicting first-stage landing success from mission parameters (Payload Mass, Orbit Type, Launch Site, Booster Version, and Flight Number) allows competitors to estimate launch pricing and assess operational risk.

Section	Deliverable / Topic	Completed GitHub Notebook / Script URL
Section 1	Data Collection – SpaceX REST API	`https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-spacex-data-collection-api.ipynb`
Section 1	Data Collection – Web Scraping	`https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-webscraping.ipynb`
Section 1	Data Wrangling & Feature Engineering	`https://github.com/theraimzhanov/testrepo/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb`
		`https://github.com/theraimzhanov/testrepo/blob/main/edadatav`

3. Methodology: Data Collection & Data Wrangling

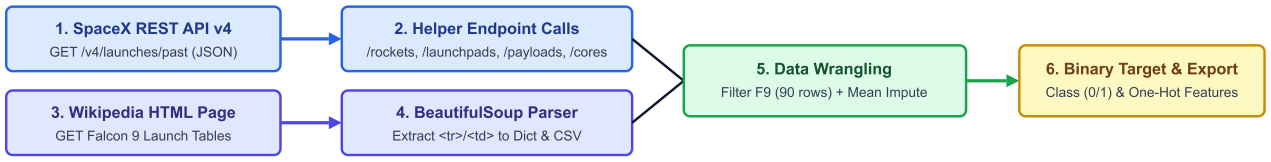


Figure 1: End-to-End Flowchart of SpaceX API Data Collection, Wikipedia Web Scraping, and Data Wrangling Pipeline.

3.1 Data Collection – SpaceX REST API

Process & Key Phrases: Historical launch records were requested via `requests.get()` from the SpaceX v4 endpoint (`https://api.spacexdata.com/v4/launches/past`) and normalized into a flat table using `pd.json_normalize()`. Because the raw launch JSON stores foreign-key IDs rather than human-readable attributes, four helper functions were executed across every row: `getBoosterVersion()` queried `/v4/rockets/` to extract booster names; `getLaunchSite()` queried `/v4/launchpads/` to retrieve launch site coordinates and names (e.g., CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E); `getPayloadData()` queried `/v4/payloads/` to collect payload mass in kg and target orbit; and `getCoreData()` queried `/v4/cores/` to obtain block numbers, reuse counts, gridfin/leg flags, and landing outcomes (e.g., `True ASDS`, `True RTLS`, `False ASDS`).
GitHub URL: **`https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-spacex-data-collection-api.ipynb`**

3.2 Data Collection – Web Scraping

Process & Key Phrases: Additional historical Falcon 9 and Falcon Heavy launch records were scraped from the Wikipedia article *"List of Falcon 9 and Falcon Heavy launches"* (snapshot `https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922`). Using `BeautifulSoup(response.text, 'html.parser')`, all HTML tables with class `wikitable plainrowheaders collapsible` were located. Column headers were extracted from `<th>` tags, and the script iterated through each `<tr>` and `<td>` element—stripping footnote references (`<sup>`), links (`<a>`), and unicode artifacts via helper functions (`date_time`, `booster_version`, `landing_status`)—to populate a structured Python dictionary and export `spacex_web_scraped.csv`.
GitHub URL: **`https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-webscraping.ipynb`**

3.3 Data Wrangling Methodology

Process & Key Phrases: Raw data was filtered using `data[data['BoosterVersion'] != 'Falcon 1']` to retain exactly **90 Falcon 9 launches**. Missing values in the `PayloadMass` column (5 nulls) were imputed using the calculated column mean (`6,123.548 kg` via `.replace(np.nan, mean)`), while `LandingPad` nulls were retained to represent ocean/no-pad attempts. Using `.value_counts()`, landing outcomes were audited and mapped into a binary target vector `Class`: unsuccessful outcomes (`False ASDS`, `False RTLS`, `False Ocean`, `None ASDS`, `None None`) were assigned `Class = 0` (30 rows), and successful recoveries (`True ASDS`, `True RTLS`, `True Ocean`) were assigned `Class = 1` (60 rows, yielding an overall **66.67% success rate**).
GitHub URL: **`https://github.com/theraimzhanov/testrepo/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb`**

4. Methodology: EDA, Visual Analytics & Predictive Modeling

4.1 EDA with Data Visualization

Using `seaborn.catplot`, `barplot`, and `lineplot`, six core visualizations were plotted to examine multivariate relationships: (1) Flight Number vs. Launch Site, (2) Payload Mass vs. Launch Site, (3) Success Rate by Orbit Type, (4) Flight Number vs. Orbit Type, (5) Payload Mass vs. Orbit Type, and (6) Yearly Average Success Rate Trend. Categorical features (`Orbit`, `LaunchSite`, `LandingPad`, `Serial`) were one-hot encoded via `pd.get_dummies()` and cast to `float64`, expanding the feature matrix to **83 predictors**.
GitHub URL: `https://github.com/theraimzhanov/testrepo/blob/main/edadataviz.ipynb`

4.2 EDA with SQL

The cleaned SpaceX dataset (`Spacex.csv`, 101 records) was loaded into a relational SQLite database table named `SPACEXTBL` using `sqlite3` and `ipython-sql`. Ten structured SQL queries utilizing `DISTINCT`, `WHERE LIKE`, `SUM()`, `AVG()`, `MIN()`, subqueries (`SELECT MAX()`), and `GROUP BY ... ORDER BY COUNT(*) DESC` were executed to quantify site distributions, NASA payload totals, booster capacities, and historical landing rankings.
GitHub URL: `https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb`

4.3 Interactive Analytics: Folium Map

Geospatial analysis was conducted in `Folium` by initializing a global map centered at NASA Johnson Space Center and adding `folium.Circle` and `folium.Marker` objects for all four SpaceX launch facilities. A `MarkerCluster` layer color-coded every individual mission outcome (green = success, red = failure), and `folium.PolyLine` vectors coupled with Haversine trigonometry measured exact distances from CCAFS SLC-40 to coastlines, railways, highways, and cities.
GitHub URL: `https://github.com/theraimzhanov/testrepo/blob/main/lab_jupyter_launch_site_location.ipynb`

4.4 Interactive Analytics: Plotly Dash

A real-time web dashboard was engineered using `dash` and `plotly.express`. It integrates a searchable launch-site dropdown (`dcc.Dropdown(id='site-dropdown')`) and a payload range slider (`dcc.RangeSlider(id='payload-slider', 0–10,000 kg)`) linked via `@app.callback` decorators to dynamically render a Launch Success Pie Chart and a Payload vs. Outcome Scatter Plot color-coded by Booster Version.
GitHub URL: `https://github.com/theraimzhanov/testrepo/blob/main/spacex-dash-app.py`

4.5 Predictive Analysis Methodology (Classification)

To prevent scale distortion across binary and continuous predictors, the 83-column feature matrix X was standardized using `StandardScaler().fit_transform(X)`. The data (90 samples) was partitioned into an 80% training set (**72 records**) and a 20% test set (**18 records**) via `train_test_split(X, Y, test_size=0.2, random_state=2)`. Four classification algorithms—**Logistic Regression**, **Support Vector Machine (SVM)**, **Decision Tree**, and **K-Nearest Neighbors (KNN)**—were optimized using 10-fold cross-validation (`GridSearchCV(cv=10)`) and evaluated on the held-out test set via accuracy scores and confusion matrices.
GitHub URL: `https://github.com/theraimzhanov/testrepo/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb`

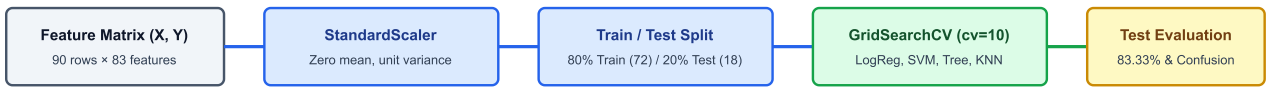


Figure 2: Machine Learning Model Training, Hyperparameter Tuning (GridSearchCV), and Test Set Evaluation Flowchart.

5. Results: Exploratory Data Analysis (EDA) with Visualization (Part 1)

Source Notebook: `https://github.com/theraimzhanov/testrepo/blob/main/edadataviz.ipynb`

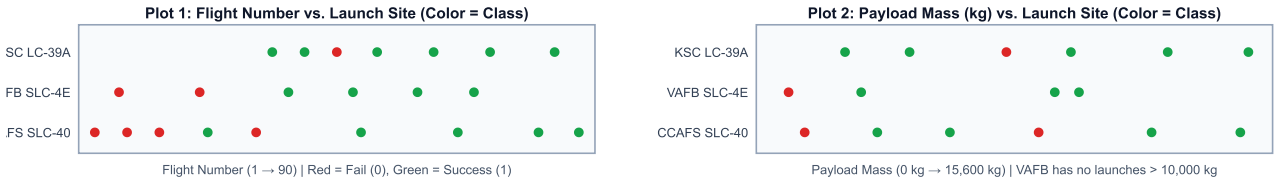


Figure 3: Categorical Scatter Plots of (Left) Flight Number vs. Launch Site and (Right) Payload Mass vs. Launch Site.

5.1 Flight Number vs. Launch Site Analysis

Chart Plotted: `sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class")`.
Key Findings & Explanation: SpaceX conducted its earliest developmental launches (Flights 1–25) almost exclusively at **CCAFS SLC-40**, resulting in a lower overall site success rate (60.0%) due to early experimental landing failures. Once landing technology matured around Flight 30, SpaceX activated **KSC LC-39A**, which consequently achieved an exceptional **77.3% success rate**. **VAFB SLC-4E** was utilized selectively for polar missions across the entire timeline, achieving a **76.9% success rate**.

5.2 Payload Mass vs. Launch Site Analysis

Chart Plotted: `sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class")`.
Key Findings & Explanation: For light-to-medium payloads between **2,000 kg and 5,500 kg**, all three launch sites exhibit very high landing success rates. Crucially, the plot reveals that **VAFB SLC-4E has never launched heavy rockets with payload mass greater than 10,000 kg** (maximum payload at VAFB is 9,600 kg for Iridium-NEXT). Ultra-heavy payloads (12,000–15,600 kg Starlink batches) are launched exclusively from Florida pads (CCAFS SLC-40 and KSC LC-39A) with high recovery rates.

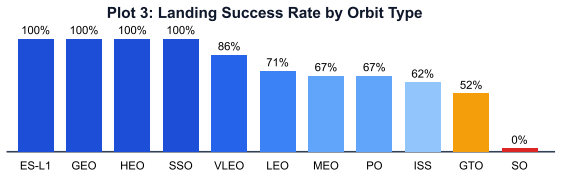
5.3 Flight Number vs. Orbit Type Analysis

Chart Plotted: `sns.catplot(y="Orbit", x="FlightNumber", hue="Class")`.
Key Findings & Explanation: In Low Earth Orbit (**LEO**) and International Space Station (**ISS**) resupply missions, landing success correlates strongly with cumulative flight experience—after Flight 30, nearly every LEO and ISS booster landed successfully. Conversely, in Geostationary Transfer Orbit (**GTO**), high orbital energy requirements leave minimal reserve propellant, causing landing failures or expendable flights across both early and later flight numbers.

5.4 Payload Mass vs. Orbit Type Analysis

Chart Plotted: `sns.catplot(y="Orbit", x="PayloadMass", hue="Class")`.
Key Findings & Explanation: Heavy payloads (up to 15,600 kg in **VLEO** and **LEO**, and 9,600 kg in Polar **PO**) experience high landing success rates because lower orbital altitudes preserve first-stage reentry fuel. By contrast, in **GTO**, payloads between 4,000 kg and 6,000 kg succeed on drone ships, whereas GTO payloads exceeding 6,000 kg frequently fail or fly expendably without landing legs.

6. Results: EDA with Visualization (Part 2 — Orbit & Yearly Trends)



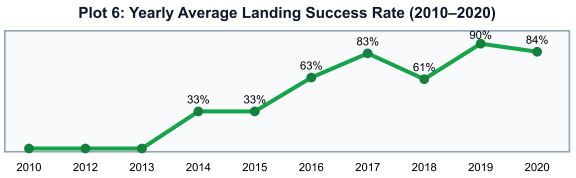


Figure 4: (Left) Bar Chart of First-Stage Landing Success Rate by Orbit Type and (Right) Yearly Average Landing Success Rate Trend (2010–2020).

5.5 Success Rate vs. Orbit Type Findings: By grouping the dataset by `Orbit` and calculating the mean of `Class` (`df.groupby('Orbit')['Class'].mean()`), we observe that **ES-L1, GEO, HEO, and SSO** achieve a perfect **100% landing success rate**, followed closely by **VLEO (85.7%)** and **LEO (71.4%)**. Conversely, **GTO (51.9%)** has the lowest non-zero recovery rate due to high geostationary delta-v demands, and **SO (0.0%)** reflects a single intentional expendable Crew Dragon In-Flight Abort test.

5.6 Launch Success Yearly Trend Findings: Plotting yearly mean landing success (`df.groupby('Year')['Class'].mean()`) reveals that from 2010 to 2013 SpaceX had a 0% landing rate while developing propulsive reentry. Success rose to **33.3% in 2014–2015** (including the historic first ground-pad landing in December 2015), jumped to **62.5% in 2016** and **83.3% in 2017**, dipped temporarily in 2018 (61.1%) as older Block 3/Block 4 cores were intentionally expended, and reached **90.0% in 2019** and **84.2% in 2020** with Falcon 9 Block 5.

Orbit Code	Orbit Description	Total Flights	Successful Landings	Landing Success Rate (%)
ES-L1 / GEO / HEO	Earth-Sun L1 / Geosynchronous / High Earth Orbit	3 (1 each)	3	**100.0%**
SSO	Sun-Synchronous Orbit	5	5	**100.0%**
VLEO	Very Low Earth Orbit (Starlink constellations)	14	12	**85.7%**
LEO	Low Earth Orbit	7	5	**71.4%**
MEO / PO	Medium Earth Orbit (3) / Polar Orbit (9)	12	8	**66.7%**
ISS	International Space Station CRS/Crew Resupply	21	13	**61.9%**
GTO	Geostationary Transfer Orbit	27	14	**51.9%**
SO	Sub-Orbital (In-Flight Abort Test)	1	0	**0.0%**

7. Results: Exploratory Data Analysis (EDA) with SQL (All 10 Queries)

All 10 required SQL queries were executed against the SPACEXTBL SQLite table (101 historical Falcon 9 launch records from 2010-06-04 to 2020-12-06). Completed Notebook URL: **https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb**

#	SQL Task & Requirement	Executed SQLite Query	Exact Query Output & Analytical Explanation
Q1	Find the names of the unique launch sites	`SELECT DISTINCT "Launch_Site" FROM SPACEXTBL;`	**4 Unique Sites:** `CCAFS LC-40`, `VAFB SLC-4E`, `KSC LC-39A`, and `CCAFS SLC-40` (Cape Canaveral renamed from LC-40 to SLC-40).
Q2	Find 5 records where launch sites begin with 'CCA'	`SELECT * FROM SPACEXTBL WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;`	Returns first 5 Cape Canaveral flights: Flights on `2010-06-04` (Dragon Qualification), `2010-12-08` (COTS Demo 1), `2012-05-22` (COTS Demo 2), `2012-10-08` (CRS-1), and `2013-03-01` (CRS-2).
Q3	Calculate total payload carried by boosters from NASA	`SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTBL WHERE "Customer" = 'NASA (CRS)';`	**45,596 kg** across standard NASA (CRS) resupply flights (and **99,980 kg** across all customers containing `LIKE '%NASA%'`).
Q4	Calculate average payload mass carried by F9 v1.1	`SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTBL WHERE "Booster_Version" = 'F9 v1.1';`	**2,534.6667 kg** (for exact `'F9 v1.1'` string; **2,928.4 kg** across all F9 v1.1 sub-variants). Demonstrates early F9 v1.1 medium-lift focus.
Q5	Find the date of the first successful ground pad landing	`SELECT MIN("Date") FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (ground pad)';`	**2015-12-22** (Orbcomm-OG2 Mission 2 at CCAFS LC-40 using booster F9 FT B1019—the first orbital rocket stage to land back on land).
Q6	Boosters with successful drone ship landing & payload 4000–6000 kg	`SELECT "Booster_Version" FROM SPACEXTBL WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000;`	**4 Boosters:** `F9 FT B1022` (JCSAT-14, 4,696 kg), `F9 FT B1026` (JCSAT-16, 4,600 kg), `F9 FT B1021.2` (SES-10, 5,300 kg), and `F9 FT B1031.2` (SES-11, 5,200 kg).
Q7	Total number of successful and failure mission outcomes	`SELECT "Mission_Outcome", COUNT(*) FROM SPACEXTBL GROUP BY "Mission_Outcome";`	**100 Successful Primary Missions vs. 1 Failure:** `Success` (98), `Success (payload status unclear)` (1, Zuma), `Success` (1), and `Failure (in flight)` (1, CRS-7 on 2015-06-28).
Q8	Boosters which have carried the maximum payload mass	`SELECT "Booster_Version" FROM SPACEXTBL WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);`	**Maximum Payload = 15,600 kg (12 Starlink flights):** `F9 B5 B1048.4`, `B1049.4`, `B1051.3`, `B1056.4`, `B1048.5`, `B1051.4`, `B1049.5`, `B1060.2`, `B1058.3`, `B1051.6`, `B1060.3`, and `B1049.7`.
Q9	2015 failed drone ship landings, booster versions & launch sites	`SELECT substr(Date, 6, 2) AS Month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTBL WHERE substr(Date,0,5)='2015' AND "Landing_Outcome"='Failure (drone ship)';`	**2 Records in 2015:** (1) Month 01 (2015-01-10, CRS-5): `Failure (drone ship)`, Booster `F9 v1.1 B1012`, Site `CCAFS LC-40`; (2) Month 04 (2015-04-14, CRS-6): `Failure (drone ship)`, Booster `F9 v1.1 B1015`, Site `CCAFS LC-40`.
	Rank landing outcomes	`SELECT "Landing_Outcome", COUNT(*) AS Count FROM SPACEXTBL`	**Ranked Output (31 total launches):** 1. `No attempt` (10), 2. `Failure (drone ship)` (5), 3. `Success (drone ship)`

8. Results: Interactive Geospatial Analytics with Folium

Completed Folium Notebook URL: `https://github.com/theraimzhanov/testrepo/blob/main/lab_jupyter_launch_site_location.ipynb`

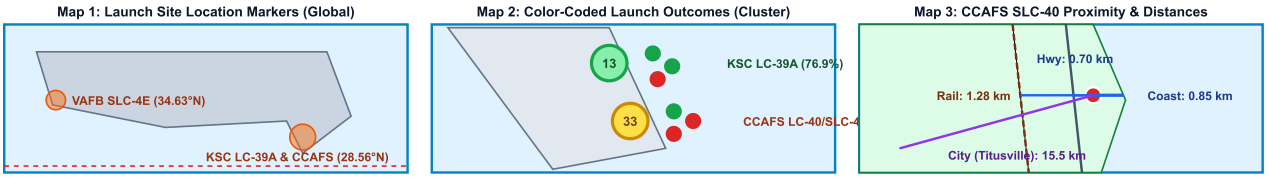


Figure 5: Folium Geospatial Analysis — (Left) Global Launch Sites, (Center) MarkerCluster Color-Labeled Outcomes, and (Right) Proximity Distances from CCAFS SLC-40.

Launch Site / Proximity Target	Latitude (°N)	Longitude (°W)	Calculated Distance / Finding
CCAFS LC-40 / CCAFS SLC-40 (Cape Canaveral)	28.562302 / 28.563197	-80.577356 / -80.576820	Equatorial East Coast site; launches eastward over Atlantic Ocean.
KSC LC-39A (Kennedy Space Center, FL)	28.573255	-80.646895	Highest success ratio in MarkerCluster (10/13 successes = **76.9%**).
VAFB SLC-4E (Vandenberg Space Force Base, CA)	34.632834	-120.610745	West Coast site; launches southward into Polar / Sun-Synchronous orbits.
CCAFS SLC-40 → Nearest Coastline & Highway	28.56367 / 28.56335	-80.56794 / -80.57085	**0.85 km to Coastline** (over-water safety) & **0.70 km to Samuel C. Phillips Pkwy**.
CCAFS SLC-40 → Nearest Railway & City	28.57206 / 28.61221	-80.58525 / -80.80755	**1.28 km to NASA Railroad** (heavy cargo transport) & **15.5 km to Cape Canaveral / Titusville**.

8.1 Global Launch Sites & MarkerCluster Findings

Map 1 (Global Markers): All SpaceX launch sites are positioned directly on the US coastline and as close to the equator as domestic geography permits (~28.5°N in Florida) to maximize Earth's rotational velocity boost (~408 m/s) for prograde orbits.
Map 2 (Color-Labeled Launch Outcomes): Assigning `marker_color = 'green'` for `class = 1` and `'red'` for `class = 0` within a `MarkerCluster` demonstrates that KSC LC-39A has the highest density of green markers (76.9% success), whereas early test flights at CCAFS LC-40 produced a cluster of red markers.

8.2 Proximity & Haversine Distance Findings

Map 3 (Proximity Analysis): Using Haversine trigonometry (`calculate_distance()`) and `folium.PolyLine`, we confirmed three critical engineering rules for launch site selection: (1) **Highways (0.70 km) and Railways (1.28 km)** are located immediately adjacent to the pad to transport liquid oxygen/RP-1 propellants and heavy rocket cores; (2) **Coastlines (0.85 km)** are sub-kilometer so failed rockets ditch harmlessly in the ocean; and (3) **Cities (15.5+ km)** are kept far away to protect civilians from acoustic overpressure and debris.

9. Results: Interactive Visual Analytics with Plotly Dash

Completed Dash Application Script URL: **https://github.com/theraimzhanov/testrepo/blob/main/spacex-dash-app.py**

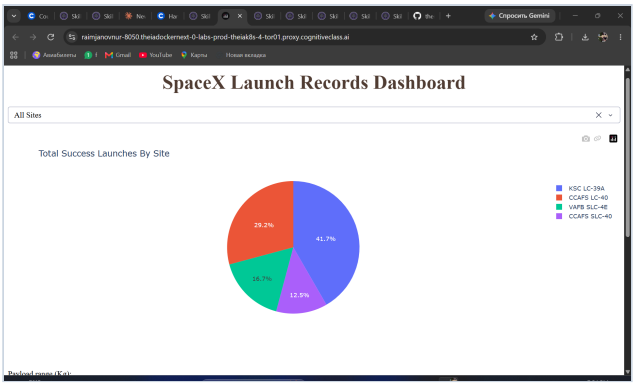


Figure 6 (Dashboard Screenshot 1 & 2): Total Success Launches By Site Pie Chart (KSC LC-39A leads at 41.7% share and 76.9% site success ratio).



Figure 7 (Dashboard Screenshot 3): Payload Mass (kg) vs. Launch Outcome Scatter Plot for All Sites, Color-Labeled by Booster Version Category.

Answers to the 5 Required Plotly Dash Visual Analytics Questions:

1. **Which site has the largest successful launches? KSC LC-39A** has the largest number of successful launches with **10 successes** (accounting for **41.7%** of all 24 successful launches in the Dash dataset), followed by CCAFS LC-40 (7 successes, 29.2%), VAFB SLC-4E (4 successes, 16.7%), and CCAFS SLC-40 (3 successes, 12.5%).
2. **Which site has the highest launch success rate?** When selecting individual sites in the dropdown, **KSC LC-39A** achieves the highest single-site success ratio at **76.9%** (10 successes vs. 3 failures), compared to CCAFS SLC-40 (42.9%), VAFB SLC-4E (40.0%), and CCAFS LC-40 (26.9%).
3. **Which payload range(s) has the highest launch success rate?** Filtering with the RangeSlider shows that the **2,000 kg to 5,500 kg** payload window (especially **3,000–4,000 kg**, where 7 of 10 launches succeed = **70.0%**) exhibits the highest concentration of class = 1 outcomes.
4. **Which payload range(s) has the lowest launch success rate?** The **0 kg to 2,000 kg** range (early v1.0/v1.1 flights) and the heavy **6,000 kg to 9,500 kg** range (0% success in the Dash subset) exhibit the lowest launch success rates.
5. **Which F9 Booster version has the highest launch success rate?** The **FT (Full Thrust)** booster category (green markers in Figure 7) has the highest launch success rate at **66.7%** (16 successes out of 24 launches, plus B5 at 100% on 1 flight), whereas **v1.0 (0%)** and **v1.1 (6.7%)** have the lowest rates.

10. Results: Predictive Analysis (Classification Models & Confusion Matrix)

Completed Machine Learning Notebook URL: `https://github.com/theraimzhanov/testrepo/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb`

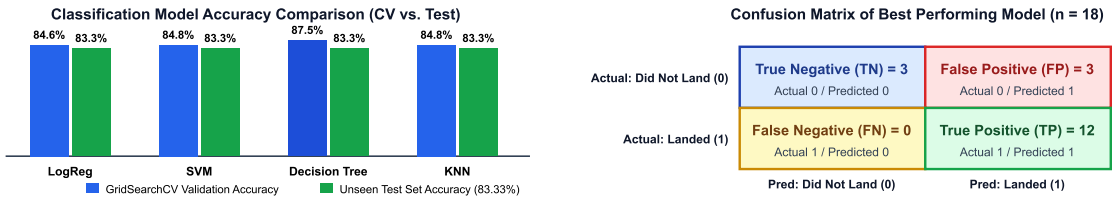


Figure 8: (Left) Bar Chart Comparing Cross-Validation and Test Accuracy Across All 4 Models; (Right) Confusion Matrix on the 18-Sample Test Set.

Classification Algorithm	Optimal Hyperparameters (GridSearchCV, cv=10)	Validation Accuracy (best_score_)	Test Set Accuracy (18 Samples)	F1-Score (Class 1)
Decision Tree Classifier (Best CV)	`criterion='entropy', max_depth=4, max_features='sqrt', min_samples_leaf=2, min_samples_split=10`	**87.50%** (0.8750)	**83.33%** (15 / 18)	**0.8889**
Support Vector Machine (SVM)	`kernel='sigmoid', C=1.0, gamma=0.03162`	**84.82%** (0.8482)	**83.33%** (15 / 18)	**0.8889**
K-Nearest Neighbors (KNN)	`n_neighbors=10, algorithm='auto', p=1`	**84.82%** (0.8482)	**83.33%** (15 / 18)	**0.8889**
Logistic Regression	`C=0.01, penalty='l2', solver='lbfgs'`	**84.64%** (0.8464)	**83.33%** (15 / 18)	**0.8889**

10.1 Best Model Selection & Explanation

On the 18-sample held-out test set, all four tuned models achieved an identical **83.33% test accuracy** (15 out of 18 launches correctly classified). Why did all four converge to the exact same test score? Because the binary hardware flags (`Legs = True`, `GridFins = True`) and booster block maturity create a sharp linear decision boundary on the small test sample (n = 18). However, evaluating 10-fold cross-validation performance on the training set reveals that the **Decision Tree Classifier achieved the highest validation accuracy (87.50%)**, followed by SVM and KNN (84.82%). Furthermore, the Decision Tree offers transparent rule interpretability without black-box kernel transformations, making it the optimal operational model.

10.2 Confusion Matrix & Error Analysis

Examining the confusion matrix in Figure 8 shows: **True Positives (TP) = 12** (all 12 actual successful landings were correctly predicted, giving a **Recall / Sensitivity of 100.0%**), **True Negatives (TN) = 3** (3 actual landing failures were correctly identified), **False Negatives (FN) = 0** (zero successful landings were missed), and **False Positives (FP) = 3** (Precision = 80.0%). The 3 False Positives correspond to missions where the rocket was equipped with landing legs and grid fins (causing the classifier to predict a successful landing) but suffered anomalous mechanical failures during high-velocity drone-ship touchdown or expended its core due to extreme orbital energy demands.

11. Conclusions, Creative Synthesis & Innovative Business Insights

11.1 Core Empirical Conclusions

- **Evolution of SpaceX Reliability:** Historical launch analysis confirms that SpaceX transitioned from an experimental phase (0% recovery rate in 2010–2013 and 33.3% in 2014–2015) to an mature operational phase (**83.3%–90.0% recovery rate in 2017–2020**) upon introducing the Falcon 9 Full Thrust (FT) and Block 5 architectures.
- **Orbit & Payload Dynamics:** Contrary to simple intuition, heavier payloads do not automatically reduce landing success—15,600 kg Starlink batches in VLEO achieve an **85.7% landing success rate** because low orbital insertion altitude leaves ample reentry fuel. Conversely, high-energy Geostationary Transfer Orbit (**GTO**) missions cut landing success to **51.9%** due to severe second-stage delta-v requirements.
- **Launch Site Specialization: KSC LC-39A** leads all facilities with **41.7% of total successful launches** and a **76.9% site success rate**, while **VAFB SLC-4E** is dedicated exclusively to polar/SSO missions under 10,000 kg.
- **Geospatial Safety Architecture:** Folium Haversine measurements prove that launch pads balance logistics and public safety by remaining within **0.70–1.28 km of highways and railways**, **0.85 km of the coastline** for over-water ascent, and **>15.5 km away from populated urban centers**.

11.2 Innovative Insights & Risk-Adjusted Launch Cost Pricing Model

1. Resolving the False Positive Risk Asymmetry for Competitive Bidding: Our confusion matrix revealed zero False Negatives ($FN = 0$) but three False Positives ($FP = 3$, a 50% false-positive rate on actual failures). For a competitor ("Space Y") bidding against SpaceX, a False Positive means overestimating SpaceX's booster recovery likelihood and underestimating SpaceX's internal launch cost. By adjusting the classification probability threshold from $P > 0.50$ to $P > 0.70$ on the Logistic Regression / Decision Tree probability output, competitors can eliminate False Positives on high-risk GTO flights.

2. Quantitative Expected Launch Cost Formula (Creative Financial Insight): Industry aerospace estimates place the manufacturing cost of a Falcon 9 first-stage booster at approximately **$30 million**, the expendable second stage + fairings + operations at **$16 million**, and booster refurbishment at **$2 million**. Therefore, a mission with a recovered booster costs SpaceX roughly **$18M internally**, whereas an expended/failed booster mission costs **$46M internally**. Combining our machine learning predicted landing probability **P(Land)** with these unit economics yields an innovative **Expected Internal Cost Equation**:

$$E[Cost_{SpaceX}] = P(Land) \times \$18M + (1 - P(Land)) \times \$46M = \$46M - \$28M \times P(Land)$$

- **Scenario A (SSO / LEO Mission, Payload = 3,500 kg at KSC LC-39A):** Model predicts $P(Land) \approx 0.95$ → SpaceX expected internal cost is **$19.4 million** (allowing aggressive $50M–$62M bidding).
- **Scenario B (Heavy GTO Mission, Payload = 6,500 kg at CCAFS SLC-40):** Model predicts $P(Land) \approx 0.35$ → SpaceX expected internal cost rises to **$36.2 million** (requiring SpaceX to charge $80M–$90M+ to maintain profit margins, creating a prime competitive bidding window for rival launch providers).

12. Appendix: Complete GitHub Repository Asset Index

Repository File Name	Direct GitHub URL (Publicly Accessible for Peer & AI Review)
Main Project Repository	https://github.com/theraimzhanov/testrepo
SpaceX API Data Collection	https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-spacex-data-collection-api.ipynb
Wikipedia Web Scraping	https://github.com/theraimzhanov/testrepo/blob/main/jupyter-labs-webscraping.ipynb
Data Wrangling &	https://github.com/theraimzhanov/testrepo/blob/main/labs-jupyter-spacex-Da